

Ning Wei

(+1)217-550-2311 | ningwei3@illinois.edu | www.linkedin.com/in/ning-wei-allen0503 | allenwn.github.io

EDUCATION

University of Illinois Urbana-Champaign | Grainger College of Engineering

Expected May 2027

Bachelor of Science in Computer Engineering

James Scholar | *Dean's List: 2024, 2025, 2026* | *GPA : 3.85/4.00*

Relevant Coursework: Data Structures, Algorithms and Models of Computation, Computer Systems Engineering, Applied Machine Learning, Deep Learning for Computer Vision, Embedded Systems, Database Systems, Digital Systems Laboratory

PUBLICATION

Junsheng Huang, Yifan Sun, Zhuoer Zhang, **Ning Wei**, Yening Liu, Michael Molter, Pavan Kumar Hanumolu, Elyse Rosenbaum, Bin Hu, and Huan Zhang. **"SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design."** *NeurIPS 2026, accepted Sep. 2026.*

RESEARCH EXPERIENCE

Assured & Trustworthy AI Research Lab (Advisor: Prof. Huan Zhang) | UIUC CS Department Champaign, IL, U.S.

Undergraduate Researcher

January 2026 - Present

- Investigated the use of large language models for analog IC design, training open-source LLMs to generate specification-conditioned SPICE netlists from natural-language circuit requirements and iteratively improve designs through simulation feedback.
- Built components of an automated simulation-grounded evaluation and optimization pipeline, integrating netlist sanity checks, Bayesian Optimization-based device sizing, circuit-specific PySpice/SPICE testbenches, performance metric extraction, and feedback construction for SFT/RL-based model refinement. Utilized simulation results and performance gaps to guide feedback-based netlist revision and improve LLM-generated circuit designs.

BioSensors Lab (Advisor: Prof. Viktor Gruev) | UIUC ECE Department

Champaign, IL, U.S.

Undergraduate Researcher

January 2026 - Present

- Contributed to an AR-assisted intraoperative tumor imaging system that uses wavelength-specific optical responses and real-time image processing to enhance visualization of tumor regions during surgical procedures.
- Developed embedded control firmware for a portable PCB-based laser illumination system using ESP32 and FreeRTOS, managing laser output, power domains, device states, user interface, safety interlocks, and illumination control for biomedical imaging experiments.
- Fabricated resin-based 3D-printed optical tissue phantoms with scattering particles and absorbing dyes to mimic biological optical properties and provide repeatable test samples for SFDI and tumor imaging system validation.

WORK EXPERIENCE

Yuanli AI Co., Ltd. (Natural Language Processing Lab)

Beijing, China

Deep Learning Algorithm Intern

June 2025 - August 2025

- Fine-tuned open-source large language base model for the education domain through continued pre-training, supervised fine-tuning, and sparse expert inference optimization, enhancing its performance in educational Q&A, problem-solving, and writing tasks.
- Developed a LLM-based k-12 math problem-solving visualization pipeline, transforming problems and solutions into structured plotting instructions, curating training data with reasoning steps and executable drawing code, and implementing a rendering engine to visualize the model's solution process.

SuperMap Software Co., Ltd.

Beijing, China

AI Remote Sensing Data Engineering Intern

May 2025 - June 2025

- Researched and analyzed state-of-the-art semantic change detection models, including Siamese networks, early fusion, late fusion, and self-attention, to evaluate their advantages and applicability.
- Reproduced representative open-source models (e.g., ChangeFormer, LeNet), built end-to-end training pipelines, and benchmarked performance across multiple datasets to support model optimization and deployment.

IQIYI Inc.

Beijing, China

Machine Learning Engineering Intern

June 2024 - August 2024

Ning Wei

(+1)217-550-2311 | ningwei3@illinois.edu | www.linkedin.com/in/ning-wei-allen0503 | allenwn.github.io

- Researched and optimized existing Text2Video models, analyzing Diffusion Transformer and Pose-Controlled Generation for skeletal pose-guided video synthesis.
- Developed pre-processing scripts using computer vision and key-point detection to generate annotated video datasets and high-quality training data for pose recognition.
- Optimized the OpenSora-based training pipeline, integrating Control-Net for precise video generation using text and skeletal pose inputs, enhancing motion control and consistency.

PROJECT HIGHLIGHTS

Time Series-Based Stock Market Prediction Model (Matlab, Python)

July 2024 - Present

- Modeled stock market trends using LSTM and Transformer architectures to capture nonlinear dynamics and long-term dependencies for improved forecasting accuracy.
- Built high-quality datasets with technical indicators and diverse training strategies to optimize prediction models and support risk-balanced trading and investment decisions.

Real-Time Control for a Self-Balancing Biped Robot (C, RTOS, PID, IMU)

January 2026 - May 2026

CS 431 Embedded Systems Course Project

- Built an embedded real-time control stack for a self-balancing biped robot, integrating IMU-based sensing, attitude estimation, filtering, periodic task scheduling, and closed-loop motor actuation under fixed timing constraints.
- Implemented PID-based feedback control to convert posture error into motor commands, coordinating sensor acquisition, state updates, control computation, and actuator output for closed-loop balance experiments.

FPGA-Based HDMI Graphics Gaming System (SystemVerilog, C, MicroBlaze)

August 2025 - December 2025

ECE 385 Digital Systems Laboratory Final Project

- Designed a real-time FPGA graphics system implementing a 640×480 HDMI pipeline with tile-based background rendering, sprite overlay, and obstacle scrolling entirely in SystemVerilog under strict video timing constraints.
- Implemented MicroBlaze-controlled game engine in C to manage world state, player movement, collision logic, USB keyboard input, interfacing with custom hardware graphics modules through AXI4-Lite memory-mapped registers.

Custom RISC-V Operating System Kernel (C, RISC-V Assembly)

August 2025 - December 2025

ECE 391 Computer Systems Engineering Course Project

- Implemented a UNIX-like kernel supporting process creation, ELF loading, system calls, and a user-level shell, with full kernel-user privilege separation on RISC-V.
- Designed virtual memory management with page tables, trap and page fault handling, and Copy-on-Write fork semantics, and integrated block device I/O (VirtIO) for persistent storage support.

Full-Stack Course & Team Matching Platform (Python, FastAPI, MySQL, React)

August 2025 - December 2025

CS411 Database System Final Project

- Built a full-stack course discovery and team-matching platform with a FastAPI REST backend, React frontend, and normalized MySQL database deployed on Google Cloud SQL.
- Implemented authentication, matching logic, transactional updates, complex relational queries, indexing, triggers, and stored procedures to support multi-condition filtering and availability-based matching.

Cell Fluorescence Image Analysis System (Bioimage Analysis, Computer Vision)

January 2026 - Present

- Built a bioimage analysis pipeline for multi-channel confocal fluorescence microscopy images, performing cell instance segmentation, fluorescence marker extraction, single-cell quantification and marker colocalization analysis.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SystemVerilog, RISC-V Assembly

AI/ML: PyTorch, LLM Fine-Tuning, Computer Vision, Diffusion Models

Software & Systems: Git, Docker, FastAPI, React, MySQL, Linux, PySpice/SPICE

Hardware & Embedded: FPGA, ESP32, FreeRTOS, PlatformIO, Altium Designer, Vivado